3. Write program demonstrates how to perform morphological analysis using the NLTK library in Python.

5]
3s

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer

# Download (run once)
nltk.download('punkt')
nltk.download('wordnet')
nltk.download('punkt_tab') # Added to address the LookupError

text = "The cats are running faster than the dogs"

tokens = word_tokenize(text)

lemmatizer = WordNetLemmatizer()
lemmas = [lemmatizer.lemmatize(word) for word in tokens]

print("Tokens:", tokens)
print("Lemmatized words:", lemmas)
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
Tokens: ['The', 'cats', 'are', 'running', 'faster', 'than', 'the', 'dogs']
Lemmatized words: ['The', 'cat', 'are', 'running', 'faster', 'than', 'the', 'dog']
```